



Semester 2 Examinations 2016/2017

Exam Code(s) 3BME1, 3BMS2, 3BPT1, 3BS9, 3FM2, 4BMS2, 4BS2
Exam(s) 3rd Arts, 3rd & 4th Science

Module Code(s) MP307
Module(s) Modelling II, Mathematical Modelling

Paper No. 1
Repeat Paper No

External Examiner(s) Professor P. Martin
Internal Examiner(s) Dr P. Piironen
Professor M. Tuite

Instructions: Marks are awarded for question 1 in Section A **and** the TWO best questions in Section B.
All questions carry the same weight.

Duration 2 hours
No. of Pages 6
Discipline(s) Applied Mathematics
Course Co-ordinator(s) Dr P. Piironen

Requirements:

Release in Exam Venue Yes ☒ No ☐

Handout None
Statistical/ Log Tables Yes
Graph Paper None
Log Graph Paper None
Other Materials Calculator

Section A - Answer all parts (a)-(d). (Each part is worth **5 Marks**.)

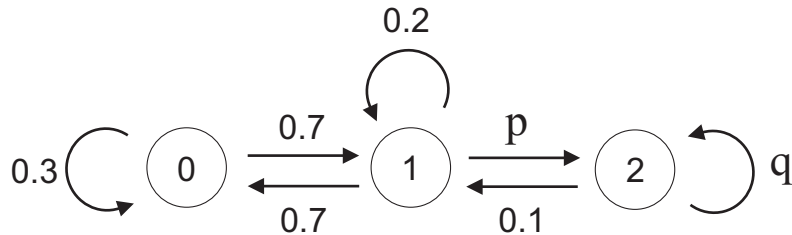
1. (a) Consider a queue and let

$$p_k = \frac{(\alpha t)^k}{k!} e^{-\alpha t}, \quad k = 0, 1, 2, \dots \quad (\text{Poisson distribution})$$

be the probability that there are k arrivals to the queue in the time interval $[0, t]$.

- (i) What is the expected number of arrivals if $t = 10$?
- (ii) What is the variance and the standard deviation of the number of arrivals if $t = 10$?

- (b) Consider the following Markov Process with transition probabilities shown.



- (i) Find the transition matrix P and the probabilities p and q .
- (ii) Is the Markov process ergodic? Explain your answer.
- (iii) Does the Markov process have unique equilibrium probabilities? Explain your answer. If the Markov process has unique equilibrium probabilities calculate them.

- (c) Consider the nonlinear population model

$$P_{n+1} = R(P_n)P_n,$$

where P_n is the population at time n and

$$R(P) = \frac{1+r}{1+r\frac{P}{K}}, \quad r, K > 0.$$

Find the fixed points for this model and show for what initial populations $P_0 \geq 0$ the population will survive as well as die out.

PTO

(d) Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= x_2 + u, \\ \frac{dx_2}{dt} &= bx_1 + x_2, \\ y &= x_1 - x_2,\end{aligned}$$

where $x_1 = x_1(t) \in \mathbb{R}$ and $x_2 = x_2(t) \in \mathbb{R}$ represent the state, $u = u(x_1, x_2) \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output and $b \in \mathbb{R}$ is the control parameter.

- (i) Is this system observable? Explain your answer.
- (ii) Is this system reachable? Explain your answer.

Section B - Answer any two questions.

2. Consider a **two-state** queue (e.g. a single telephone line) with arrival and service patterns described by Poisson processes with parameters α and β , respectively. Let

$$p_k(t) = \text{Prob}(k \text{ customers queuing at time } t), \quad k = 0, 1.$$

- (a) Sketch a diagram showing all possible transitions.

2 Marks

- (b) Show that the probabilities $p_0(t)$ and $p_1(t)$ evolve according to the differential equations

$$\begin{aligned} \frac{dp_0(t)}{dt} &= -\alpha p_0(t) + \beta p_1(t), \\ \frac{dp_1(t)}{dt} &= \alpha p_0(t) - \beta p_1(t). \end{aligned}$$

8 Marks

- (c) Calculate the equilibrium probabilities π_0 and π_1 for the two-state queue.

4 Marks

- (d) Consider a single telephone line and let 2 minutes be the average time for a call to arrive and let the average length of a call be 3 minutes. What is the probability that the phone line is busy?

6 Marks

3. Consider the life cycle of the Pacific salmon.

- (a) Let P_n be the number of mature salmon at the end of n^{th} spawning cycle. Let $Q(t)$ be the number of larvae at time t . Assume that the larvae decline according to the differential equation

$$\frac{dQ(t)}{dt} = -\beta Q(t)P_n, \quad \beta > 0, \quad Q(0) = Q_0, \quad (1)$$

where Q_0 is the number of larvae produced at the start of the $(n+1)^{st}$ cycle so that

$$Q_0 = \alpha P_n, \quad \alpha > 0. \quad (2)$$

Assume also that T is the duration of one cycle and

$$P_{n+1} = \gamma Q(T), \quad \gamma > 0. \quad (3)$$

Use equations (1)-(3) to derive the Ricker model

$$P_{n+1} = R_0 P_n e^{-\frac{P_n}{K}}, \quad (4)$$

where

$$R_0 = \alpha\gamma, \quad K = \frac{1}{\beta T}.$$

4 Marks

- (b) Use the Ricker model in (a) and determine the values of R_0 for which the salmon population P will die out.

4 Marks

- (c) Use the Ricker model in (a) and determine the values of R_0 for which a fixed salmon population P exists and determine the value of the fixed point.

4 Marks

- (d) Determine the values of R_0 for which the fixed point found in (c) is stable.

8 Marks

4. A two-compartment control system for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} u, \quad y = c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, u determines the flow rate of the drug into compartments 1 and 2 and y is the concentration of the drug in compartment 2.

(a) Investigate if the system is observable and/or reachable.

8 Marks

(b) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $y = r = 2$ and the closed loop eigenvalues for the system are $\lambda_1 = -2$ and $\lambda_2 = -2$.

12 Marks